MIL-C-51067B
24 August 1983
SUPERSEDING
MIL-C-51067A
31 August 1976

MILITARY SPECIFICATION

CALCIUM MAGNESIUM CARBONATE, TECHNICAL (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one grade of calcium magnesium carbonate, $CaMg(CO_3)_2$, known as dolomite.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1168 - Ammunition Lot Numbering

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6810

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applciable.

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

ASTM STANDARDS

C25 - Chemical Analysis of Limestone, Quicklime, and Hydrated Lime
D1193 - Reagent Water
D3951 - Commercial Packaging
E11 - Wire-Cloth Sieves For Testing Purposes

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Chemical and physical characteristics. Calcium magnesium carbonate shall conform to the chemical and physical characteristics of table I when tested as specified therein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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TABLE I. Chemical and physical characteristics

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Calcium and magnesium carbonates, moisture-free	97.0	---	4.2.4.1
Calcium carbonate, moisture-free	53	58	4.2.4.1
Moisture	---	0.5	4.2.4.1
Particle size:			4.2.4.2
Material passing a 75-micrometer (No. 200) sieve	---	5.0	
Material retained on a 180-micrometer (No. 80) sieve	---	1.0	

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the calcium magnesium carbonate produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105, inspection level S-2.

4.2.2.2 For test. Sampling shall be conducted in accordance with table II. A representative specimen of approximately 125 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 4.0 percent defective:

(a) Contents per container not as specified
(b) Container damaged or leaking
(c) Unitization not as specified
(d) Marking incorrect, missing, or illegible

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TABLE II. Sampling for test

Number of containers in batch or lot	Number of sample containers
2 to 25	2
26 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Calcium and magnesium carbonates, calcium carbonate, and moisture. Determine in accordance with the applicable portions of ASTM C25. Calculate the calcium carbonate and magnesium carbonate content based on the calcium oxide and magnesium oxide determinations in ASTM C25.

4.2.4.2 Particle size. Use sieves conforming to ASTM E11. Nest a 180-micrometer sieve on a 75-micrometer sieve equipped with a bottom pan. Transfer 100 g of the specimen to the top sieve and fasten the sieve assembly in a mechanical sieve shaker which gives 300 $\pm$ 20 gyrations and 150 $\pm$ 10 taps per minute. Operate the shaker for 5 minutes. Weigh the material in the top sieve and in the bottom pan and calculate the percent by weight specimen passing the 75-micrometer sieve and the percent by weight specimen retained on the 180-micrometer sieve.

5. PACKAGING

5.1 Unit packing, industrial. A specified quantity (see 6.2) of calcium magnesium carbonate shall be unit packed in accordance with ASTM D3951.

5.2 Packing, industrial. Calcium magnesium carbonate shall be packed in accordance with ASTM D3951.

5.2.1 Unitized loads. Uniform quantities of packs of calcium magnesium carbonate shall be unitized in a manner to assure acceptance by common carrier, stability of stacking in transit and storage, and in accordance with Uniform Freight Classification Rules, National Motor Freight Classification Rules and the rules and regulations of any other intended mode of transportation.

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5.3 Marking. Containers shall be marked in accordance with ASTM D3951. Unitized loads shall be marked to assure proper identification and timely delivery to the intended destination.

6. NOTES

6.1 Intended use. Calcium magnesium carbonate is intended for use as a retarder in grenade mixtures.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification, and the quantity of calcium magnesium carbonate to be unit packed (see 5.1).

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purposed of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

Custodians:

Army - EA
Navy - OS

Review activity:

Army - MD
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-51067B	CALCIUM MAGNESIUM CARBONATE, TECHNICAL (METRIC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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